DATA ENHANCING EXTRA

From support text to BI-ready data

Runnable lab solution • 10 October 2026

Delivery status: implemented pipeline + executed offline demo. The two instructor CSVs were not supplied. The included results use 81 fictional energy-app tickets and a rule-based adapter. OpenAI extraction, instructor-ground-truth validation, live LangSmith traces and GitHub submission remain pending.

81 tickets	20 calibration	10 consistency
All original rows preserved	At least 3 per category	Two fresh invocations each
81 schema-valid outputs	16/20 category correct	10/10 agree under rules

Demo snapshot

Measure	Observed demo result	Meaning
Category accuracy	80% (16/20)	Does not pass strict >80% target
Sentiment / urgency	95% / 75%	Measured separately on synthetic labels
Review / BI-eligible	15 / 66	Automated gate; not human approval
Provider calls / tokens / cost	0 / 0 / $0	Offline mode; no live provider measurement

What the package delivers

Python CLI, executed notebook, strict five-field schema and prompt, resilient checkpoints, category/sentiment/urgency calibration, independent consistency checks, a review queue, BI subset, README, required root summary, tests and this report. The structured CSV uses the required filename **tickets_structured.csv**.

The course requires a dedicated GitHub repository URL. A download package is the working material; it is not the final course submission. See the final page for completion steps.

1 / Explore the data and define the schema

Synthetic input: 81 unique ticket IDs spanning 1 January–31 December 2025. Text length is 3–84 characters. The CLI prints five complete fixed-seed random tickets plus the shortest and longest examples. These dates, lengths and mix describe this demo only.

Field	Successful values / contract	Purpose
category	billing, technical, feature_request, praise, other	Dominant intent / routing
sentiment	positive, neutral, negative	Expressed customer attitude
urgency	high, medium, low	Conservative text-based priority
topic	One short factual sentence	Search / drilldown context
confidence	high, low	Qualitative review signal

Prompt decisions

The prompt defines the task, untrusted ticket input, criteria and JSON output. For mixed tickets, select the dominant actionable request. A concrete fault outranks a secondary suggestion; an unclear mixture lowers confidence. Unknown text still receives best guesses and an honest low-confidence signal. Billing is not automatically negative, except the lab’s duplicate-charge convention.

Do not invent a refund request, transaction amount, deadline, root cause, customer identity, or verified incident. A topic should summarize facts in the ticket, not plausible consequences. Timestamp-derived month is computed in Python.

Example	Expected handling
I've been charged twice this month.	billing / negative / high; summarize duplicate charge without inventing a refund request
Dashboard fails; dark mode would be nice.	technical is dominant; mixed text merits review
???	other / neutral / low urgency; low confidence; no invented issue

A human can often infer intent and attitude from the words. The text alone cannot establish whether a claim is true, actual business impact, an SLA, resolution time or lifetime customer value. Confidence does not fill those information gaps.

2 / Extract robustly and trace the live run

Live mode: OpenAI Chat Completions uses a strict JSON Schema with exactly five required properties and no extra keys. The configurable default is gpt-4o-mini. The OpenAI client is wrapped with **wrap_openai**; the dictionary and record extraction interfaces use **@traceable**. Ticket input is separate from the system instructions.

Protection	Behavior
Response validation	Parse JSON; validate keys, enum values and a nonempty topic. Strict schema does not guarantee semantic correctness.
Failure handling	Refusal, truncation, parse/schema errors and API failures set all five fields to parse_error; status/error type remain separate metadata.
Transient retries	429, server, connection or timeout failures: at most 3 attempts per invocation. Other errors are not retried.
Batch preservation	Append each completed row to JSONL, flush and fsync before the next ticket; continue after a failed ticket.
Resume safety	Match input hash and prompt/schema/model/mode fingerprint. Reuse successes; optionally retry failures. Changed inputs/configurations are rejected.
Progress	Print every 10 tickets and final success/failure counts.

Live configuration and evidence

Copy .env.example to .env; provide OpenAI and LangSmith keys locally. The brief uses the EU LangSmith endpoint and project m5-ai-data-enhancing; match the endpoint to your account. Keep credentials out of GitHub. In demo mode tracing is disabled and the provider client is never created.

A fresh full live run has **104 logical invocations**: 3 sample tests + 81 batch tickets + 20 consistency invocations. Calibration reuses batch outputs. Retries may add attempts; resume still reruns sample/consistency tests. Token counts are observed usage, not necessarily all billable retry usage. Cost remains unknown unless current rates and complete usage are available.

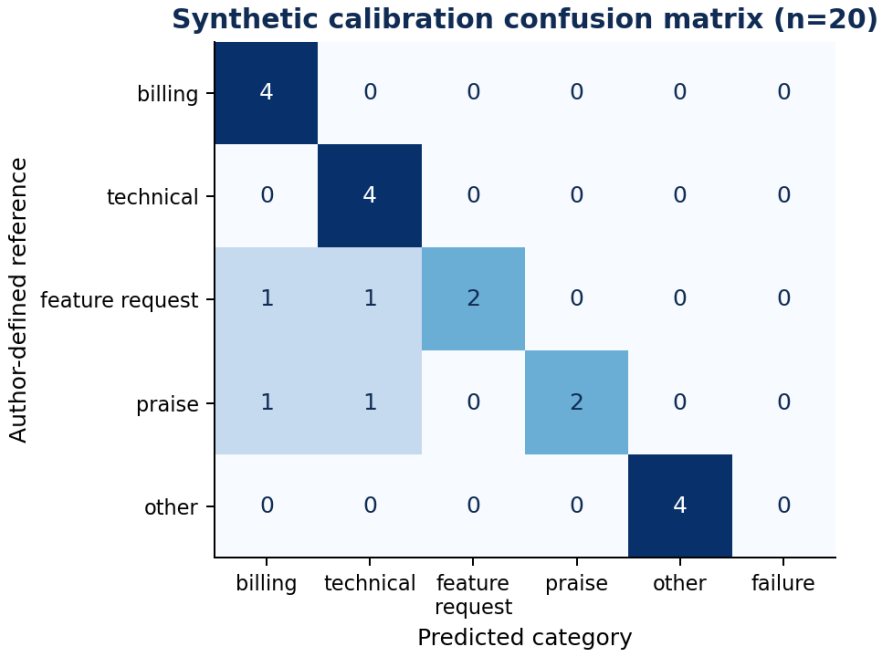
After a live run inspect actual LangSmith chains and OpenAI children for correct inputs, outputs, model, tokens, latency and available cost. Record run links and permitted screenshots. This package contains no verified live trace evidence.

3 / Validate meaning, then measure repeatability

Calibration is selected before extraction from human labels: at least three examples per category, very short and declared ambiguous cases prioritized, then filled to 20. Demo selection has four per category. It never selects examples because their predictions were correct. All20 remain in the accuracy denominator; missing or failed predictions are wrong.

Field	Correct / total	Accuracy
Category	16 / 20	80%
Sentiment	19 / 20	95%
Urgency	15 / 20	75%

Strict acceptance: >80% means 17/20 or more. The demo category score does not pass. Moreover, synthetic rules and synthetic labels do not validate the required LLM extraction even if a demo score is high.



Rules confuse feature suggestions mentioning invoices and praise about resolved faults with actionable billing/technical requests. Review raw text with the comparison file; adjudicate disputed labels and use a new held-out set after prompt changes. Balanced calibration is not a production-prevalence estimate.

Consistency: 10 tickets, two fresh invocations each

Overall demo agreement is 10/10 (100%). Both-high cohort: 4/4; any-low cohort: 6/6; failed pairs: 0. Failed pairs count against overall agreement, and empty cohorts produce null. Deterministic rules can repeat a wrong answer. Temperature zero in a live LLM run is not a guarantee of determinism; evaluate actual paired outputs.

4 / Hand off the data without hiding uncertainty

The master retains original ID, timestamp and raw text, plus the five extraction fields and audit metadata. Month is UTC YYYY-MM. **review_required** is true for low confidence or failure; **bi_eligible** is its inverse. Eligibility is an automated flag, not proof of correctness or human approval.

CSV	Rows	Use
tickets_structured.csv	81	Complete master; filter BI-eligible for routine charts
tickets_review_queue.csv	15	Human review with original text and proposed labels
tickets_bi_ready.csv	66	Provisional filtered BI subset

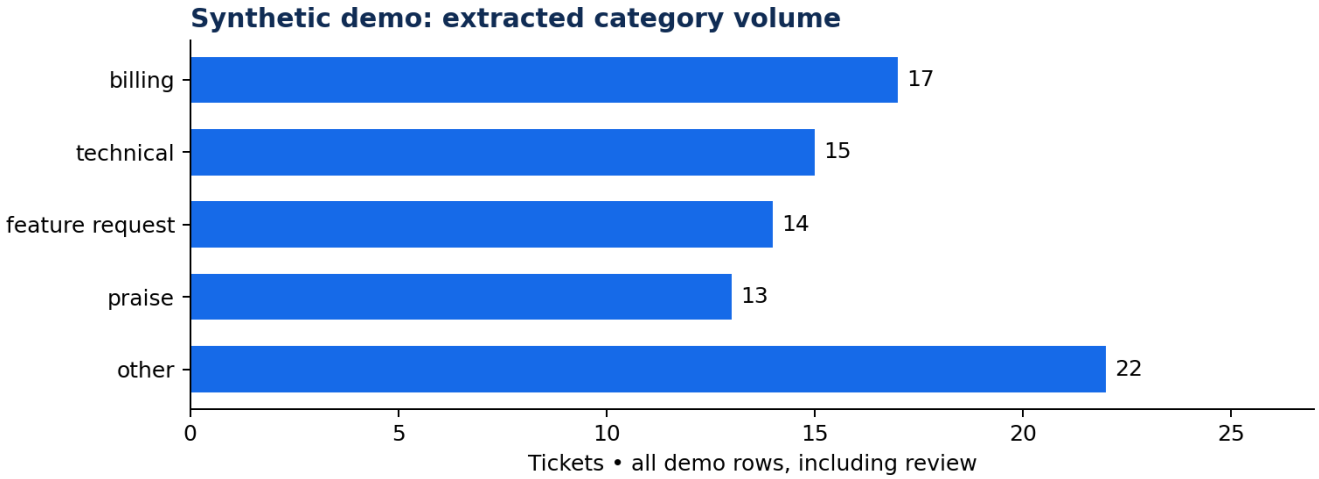


Tableau / Power BI

Import IDs as text, submitted_at as datetime, month as YYYY-MM text and review flags as booleans. Count tickets by category, compare sentiment by category and show monthly volume. Filter bi_eligible=true; never plot parse_error as a real category. Display excluded/review counts separately because exclusion can bias the charts.

Keep topic in a searchable detail table, tooltip or drilldown. It is narrative text, not a measure to sum or average. A production review flow should record corrected labels, reviewer, approval time and audit history before reinstating a row.

Human judgment and economics

Human labels can disagree on dominant intent or implied urgency. Use an annotation guide and adjudication. Review low confidence and sample high confidence for drift. Provider cost is only one part of total cost: add review, rework, setup and infrastructure. The notes include a break-even formula; live token rates and measured costs were not available here.

These category volumes reflect fictional ticket construction and rule errors. They support testing the handoff, not conclusions about a real company or energy service.

5 / Run, complete, and submit the actual lab

Step	Required action
1 · Demo	Install requirements from the repository root; run python -m src.run_lab --mode demo. Run python -m unittest discover -s tests -v.
2 · Instructor inputs	Add the two course CSVs under data/instructor. Ensure matching IDs, labels and timestamps; identify human-reviewed ambiguous examples.
3 · Live extraction	Configure .env and run python -m src.run_lab --mode live. Check failures, review queue and actual trace evidence; resume explicitly if needed.
4 · Acceptance	Verify 81 actual successful extractions, calibration category >=17/20, separate sentiment/urgency accuracies, and fresh paired consistency results.
5 · Required outputs	Live run rewrites root tickets_structured.csv and the one-paragraph lab_summary.md. Include code, run instructions and evidence. Demo PDF/HTML remain demo reports.
6 · GitHub	Create a dedicated lab-only repository, inspect staged files and sharing permissions, publish it yourself and submit its real repository URL.

Package map

src/ implements the pipeline and CLI; **notebooks/** contains the executed demo notebook; **prompts/** exposes the schema and prompt; **outputs/demo/** preserves audit results; **docs/** covers requirements, reflection, data definitions and submission; **tests/** verifies evaluation and failure invariants. README.md is the start point.

Verification performed

The offline pipeline ran end to end; master + BI subset + review queue reconcile to 81. Sixteen automated offline tests check parse/schema errors, API contract via mocks, bounded retry, failures, denominators, strict threshold, fresh paired invocations and safe checkpoint resume. Notebook execution details are recorded in docs/verification.md. No paid provider call or GitHub publication occurred.

Sources and further reading

Requirements: supplied LAB | Data Enhancing Extra.pdf (not redistributed).

OpenAI Structured Outputs

LangSmith: trace OpenAI

LangSmith: annotate code

LangSmith observability quickstart

Documentation checked 10 October 2026. See the README for full URLs and configurable model/rates.